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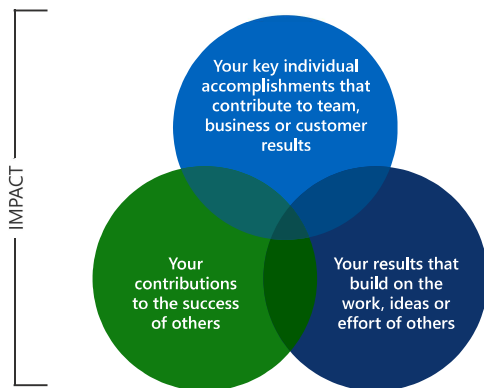
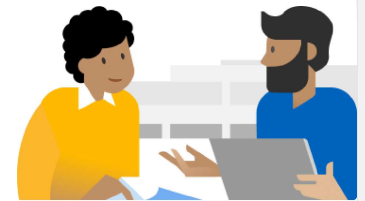
## DEVESH KUMAR

SOFTWARE ENGINEER II | Reports to: Suma Kollipara

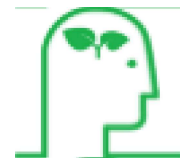
### Connect Nov 2024

Reflection Period: May 12, 2024 - May 24, 2024

Connects provide a regular touchpoint where you and your manager can come together with a growth mindset to reflect on impact and discuss areas for improvement and future development.



**Fixed Mindset** leads to a desire to look smart and therefore a tendency to avoid challenges, give up easily, see failures as fruitless or worse, ignore useful negative feedback and feel threatened by the success of others.



**Growth Mindset** leads to a desire to learn and therefore a tendency to embrace challenges, persist in the face of setback, see failures as essential to mastery, learn from criticism and find lessons and inspiration in the success of others.

### Reflect on the past

One aspect of successful impact is when you consistently deliver against all and likely exceed some expectations for your core priorities. Capture what you've accomplished and learned since your last Connect.

#### Summarize your impact

#### Core priorities

What impact did you have for each of your core priorities?

- **Copilot changes and scale motion**
  - Copilot V3

- We integrated the Copilot V3 pipeline into our codebase. This integration has allowed us to achieve an overall efficiency of more than 75% on a regular basis
- It enables customers to guide the Copilot in making queries related to their specific fields by using examples and instructions.
- **Example API Implementation**
  - Implemented CRUD APIs to enable users to register queries, which are then used in prompts by Copilot.
  - Updated the WAF policy to allow KQL written by customers to pass through the Application Gateway.
  - Added a List API to enable users to add multiple examples at once.
  - Added an API to retrieve all examples linked to an instruction.
- **Inbuilt Jobs**
  - Wrote an inbuilt alias job to register out-of-the-box aliases. These aliases are used in Copilot prompts for getting context and connecting queries with entities.
  - Wrote an inbuilt instruction job to register inbuilt instructions. Ensured that if a customer edits an existing instruction, it would not be updated.
  - Wrote an inbuilt examples job to register inbuilt examples.
- **Job to create Graph functions**
  - Developed a Graph function in ADX that abstracts graph creation using property and relationship tables.
  - Integrated Shuffle strategy in the query to make computation fast.
  - Registered these functions as startup jobs, which are utilized by Copilot in its queries.
- **Instruction Retrieval/ Prompt generation**
  - Wrote code to integrate examples, instructions and alias dictionary into prompt.
  - Added retries in the query API flow, ensuring that previous errors are propagated in the next prompt for query generation.
  - Optimized the flow by implementing parallel calls for data fetching from Cosmos.
- **Code for converting KQL to cipher text**
  - In Copilot prompts, cipher-based queries are easier for Copilot to handle compared to KQL. Made code changes to internally convert example KQL provided by users to cipher-based KQL.
  - This will allowing customers to continue using their existing KQL without needing to learn new cipher-based queries.
- **Added typecast in queries**
  - Upon investigation, it was found that Copilot often fails to add typecasting of columns in its queries, resulting in failing queries and reduced accuracy.
  - Made code changes to apply typecasting on generated queries based on the column type defined in the DTDL.
- **Fixing L2 tests**
  - Registered custom examples and queries that are required by validate API I2 test to run on bakery dataset
  - These examples and queries are specific to bakery dataset and can not be included out of the box
- **Removed V2 implementation**
  - Removed obsolete V2 APIs from the codebase for better readability and code maintainability
- **GSI and customer engagement**

- **OOE query for Parsec**
  - Wrote Overall Equipment Effectiveness Query for Parsec data that was mapped to our entities.
- **Fixing Deployment failure for Schaeffler**
  - While being DRI Schaeffler was facing issues with deployment as they did not gave correct permissions at subscription level. Worked with Shubham to fix it.
- **POC work**
  - **Assistant API spikes**
    - Created functions in ADX to denormalize the entity structure. These view functions will be used by Assistant APIs to generate data for computation.
    - Extended the Assistant API spike to include new views by writing a Python script to analyze the accuracy of Assistant APIs with the newly created views.
    - Explored and demonstrated a working solution for generating and updating graphs through the Assistant based on results generated by Copilot.
  - **Decomposition of Entity**
    - Copilot often jumbles when we have to segregate an entity into multiple parts and then apply filters over it.
    - To overcome this we tried decomposition of entity, this is like making a new entity out of an existing entity just for the copilot calls.
    - through this we were able to get similar accuracy with 50% less instruction (6 out 12) and only 3 examples (earlier it was 8)
- **Bug Fixes**
  - **Added previous validation error in prompt**
    - After the initial implementation the prompt was not getting the previous validation errors that happened on last KQL generation, Made sure that those are propagated in next run
  - **Added fix for intent detection skill**
    - Upon moving to new version of open AI the intent detection call result were adding some additional text which needed to be removed for proper classification
  - **Fix instruction id**
    - There was a mismatch in they the id field was stored in ADX and fetched in the code during retrieval.
  - **Validate doc creation failures**
    - Added await while validate doc creation call. This will fail the validate API call
  - **Responsible AI testing**
    - Tested responsible AI and jail breaking scenarios on our solution
    - Able to jail break the solution by disabling the out of the box example and instruction and registering custom instruction with malicious content
    - Actively contributed for testing the RAI Review of the solution. Added multiple dataset for the summarization part and normal flow

### What impact did your actions have in contributing to a more diverse and inclusive Microsoft?

1. Seek diverse perspectives and incorporate feedback, apply insights from a variety of sources and people.
2. Mentor and/or support a person who may benefit

### Security Core Priority

What impact did your actions have in contributing to a more secure Microsoft? You can still capture progress even before you set your Security Core Priority for the first time.

- Actively tracked Security wave 1 and 2 work items.
- Did bugbash to test different scenarios of MRG v2 changes, on how it could be manipulated by malicious entity.
- Did Responsible AI tests on our system. Found out vulnerabilities that could be potential threat to our system.



#### Manager comments

Acknowledged on the impact called out and overall, its a great year for Devesh, where he transitioned from MDS to Copilot workstream at a critical juncture and seamlessly integrated with the team and played a pivotal role in hardening the accuracy and performance improvements, which became quite important for PuP readiness. Devesh's velocity in delivery and his quick turnaround time in bringing tasks to completion with quality is top-notch. Devesh is seasoned developer and delivers work with very minimal guidance. Always willing to accept challenges and also positioned himself as core team member in the team. Would like to see Devesh's contributions in an expanded fashion across the product where he actively reviews and validates code bits from other workstreams and becoming an SME for the overall product including the MDS Analytics space in next semester.

### Reflect on a challenge or setback

In addition to your accomplishments relative to your core priorities, you must demonstrate a growth mindset to deliver successful impact. Share how you demonstrated a growth mindset by taking personal accountability for setbacks, asking for feedback, and applying learnings to have greater impact.

Consider when you could have done something differently. How will you apply what you learned to make an even greater impact?

- The POC on decomposition of API could not be replicated to code as splitting entity into two would confuse the user and hence would lower the adoption.

- For Parsec engagement the data mapping was complicated that made the OEE query to difficult for Copilot to reproduced. If mapping would be simplified then eventually query for different use cases would be simpler hence higher accuracy



## Manager comments

Aligned on the setbacks, but these are good learning experiences for developers to wear the hats of end users and overall envision the product from an end user perspective and build products with sticky adoption patterns.

## Plan for the future

Think about opportunities for impact in the upcoming Connect period. What are your 3-5 core priorities?

### Review your core priorities

#### Core priorities

##### 1. Assistant APIs

Focus areas for the upcoming period to drive business impact:

- Implementation of Assistant API pipeline in C# codebase
- Integrate views in pipeline V3 to make the query simpler and hence increasing accuracy
- Ability for users to use code interpreter of assistant Api to better visualize the results in graph from

Success measures:

- Greater customer adoption and increase in the share of users using azure studio

##### 2. Copilot Studio integration

Focus areas for the upcoming period to drive business impact:

- Focus on increasing the usage of the azure studio
- Integrating work done on assistant with azure studio

Success measures:

Greater customer adoption and increase in the share of users using azure studio

##### 3. DRI core priority

Focus areas for the upcoming period to drive business impact:

- As our product moves to public preview, there will be a lot of customer POCs and engagement. I will actively try to take part in the engagement
- I will try to resolve problems faced by customers in adopting our product.

**Success measures:**

Smooth onboarding of customer

4. **MDS core and analytics convergence**

**Focus areas for the upcoming period to drive business impact:**

- As we move forward, we will try to integrate the analytics part that the SKG team is building on Fabric with the native core solution.
- I will also try to integrate the Copilot capabilities in MDS analytics

**Success measures:**

- Unified product and increase in fabric users.
- Increase in customer ingestion as spark jobs are natively faster.

## Diversity & Inclusion (D&I) Core Priority

To realize our mission to empower every person and organization on the planet to achieve more, each and every one of us must help create a diverse and inclusive culture where everyone can bring their full and authentic self, in alignment with our values. Consider how you will **deliver successful impact by consistently demonstrating Microsoft's cultural expectations and values**. Visit the [Diversity & Inclusion \(D&I\) Core Priority page](#) for guidance and examples of impact.

Select and commit to at least one focus area from the options below, or create one of your own.

- Seek diverse perspectives and incorporate feedback, apply insights from a variety of sources and people
- Mentor and/or support a person who may benefit

**How will you define success as you activate your D&I Core Priority?**

- The perspective and feedback from others would be a good measure for success.

## Security Core Priority

We all have the responsibility to directly contribute to the security of Microsoft and our customers. We must all commit to prioritizing the design, build and delivery of technology, tools, data and processes that meet the highest possible standard of security. **For more information on the Security Core Priority, visit [HRweb](#).**

With this priority, I personally commit to being an active and intentional participant in creating a secure Microsoft by:

- Completing and applying learnings from required Security Foundations and Trust Code (Standards of Business Conduct) trainings.
- Prioritizing security in my own work and ensuring I have anticipated and implemented secure approaches to products, processes and tools **in alignment with our [Secure Future Initiative](#)**.
- Committing to leveraging [MSProtect](#) to learn more about how I can contribute to securing Microsoft, including reporting any digital or physical security incidents via [Report It Now](#).



### Manager comments

Acknowledged on the core priorities set for the upcoming semester.

## Identify your opportunities for growth

We are all expected to embrace a growth mindset and focus on continuous learning to deliver successful impact.

**What 1-2 actions will you take to focus on learning, growth, and development over the next period?**

- I will work more on the new technologies and methods coming up on assistant and copilots
- I will learn more on security aspects of our solution.

**What support do you need to fulfill these actions? Fill in any that apply.**

**Opportunities to (optional):**

No Response

**Perspective or coaching on (optional):**

No Response

**Learning on (optional):**

No Response



### Manager comments

Ack, would love to see more mentoring as well to the junior developers.

## Conversation starters (optional)

Choose any of these optional prompts to continue great conversations with your manager. Write a brief response (1-15 words).

**Of my strengths, the ones I'd like to apply more in my work are:**

**In my work, I feel motivated or energized by:**

**In my work, I lose motivation or energy from:**

**I know I can't do everything on my own. I'd like help with:**

It can be challenging to prioritize important topics for 1:1 conversations. Plan ahead and flag any of the following topics you'd like to discuss with your manager in upcoming conversations.

**Career:** Discuss ways to grow my career to align with my strengths, interest & aspirations.

**Recognition:** Discuss my [recognition preferences](#) and highlight my contributions.

**Opportunities:** Increase my responsibilities, such as new core priorities, stretch assignments, promotion or becoming a people manager.

**Alignment:** Clarify priorities and expectations for successful impact.

**Learning:** Explore potential development and growth opportunities to optimize your performance.

**Feedback:** Get input on how I'm doing in my role and how to deliver great impact.

**Flexibility:** Consider how you work best and discuss your workplace preferences (i.e. site, location, hours)

**Wellbeing:** Discuss any dimensions of being well: mental, emotional, physical, financial.

Check out [aka.ms/career](https://aka.ms/career) for resources and guides on developing your career.



## Manager comments